

PRIYANKA DESHPANDE – Sr. Analytical Engineer

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PROFESSIONAL SUMMARY

- 8+ years of experience spanning Data Engineering, Analytics, and Data Science, building SQL-driven analytical data models and end-to-end pipelines that turn raw, unstructured data into decision-ready assets.
- Deep hands-on SQL expertise across Synapse Serverless SQL Pools, Azure SQL Database, SQL Server, PostgreSQL, and MySQL, including query optimization, external table design, and building fact/dimension schemas for scalable reporting.
- Advanced Tableau developer, translating model output and pipeline data into interactive, stakeholder-facing dashboards covering complaint trends, phishing risk, and document-processing KPIs; also proficient in Power BI.
- Built and scheduled Apache Airflow DAGs to orchestrate ingestion and processing workflows for a production AI/document-intelligence pipeline, in addition to Azure Data Factory (ADF) pipeline orchestration.
- Healthcare domain experience from a data science role at Evolent (healthcare services), plus enterprise Azure data platform delivery (ADF, Synapse Analytics, Databricks, ADLS Gen2) using Python, SQL, and PySpark.
- Certified Scrum Master who partners closely with Product and Engineering teams, running sprint ceremonies and removing delivery blockers while owning technical execution end-to-end.
- Familiar with dbt for developing SQL-based ELT transformations, modular data models, testing, and documentation to support scalable analytics workflows.
- Familiar with Starburst/Trino concepts for distributed SQL query execution and federated analytics across multiple data sources.
- Guided junior engineers through SQL development, data modeling best practices, and dashboard design while supporting technical knowledge sharing across the team.

TECHNICAL SKILLS

SQL & Databases	Expert SQL, Azure SQL Database, SQL Server, PostgreSQL, MySQL, MongoDB, Synapse Serverless SQL Pools, query optimization, external tables
Analytics & BI	Advanced Tableau, Power BI, Matplotlib, Seaborn, Plotly, dashboard design, stakeholder reporting
Orchestration	Apache Airflow (DAG development & scheduling), Azure Data Factory (ADF), Azure DevOps CI/CD
Data Engineering	ETL/ELT pipelines, data modeling (fact/dimension, star/snowflake schema), Parquet, Delta Lake, data quality & governance, Unity Catalog, batch & streaming pipelines
Programming	Python, SQL, PySpark, Pandas, NumPy
Azure Platform	Azure Databricks, Azure Synapse Analytics, ADLS Gen2, Azure Key Vault, Azure Monitor, Log Analytics, Docker
ML / GenAI	Scikit-learn, XGBoost, MLflow, MLOps, LLM/RAG, LangChain, FastAPI, Azure OpenAI Service
Version Control	GitHub, Git
Domain	Healthcare (Evolent), Automotive/Consumer Complaints, Financial/KYC Documents
Methodology	Agile/Scrum, Certified Scrum Master

PROFESSIONAL EXPERIENCE

TD Bank, Ewing Township, NJ

Oct 2023 - PRESENT

Senior Machine Learning / AI Engineer

Project: AI-Powered KYC Document Intelligence & Verification System | Tech Stack: Azure Databricks, PySpark, Python, Apache Airflow, Azure Data Factory, LLM, MLflow, MLOps, Agile/Scrum

- Built and scheduled Apache Airflow DAGs to orchestrate the end-to-end KYC pipeline — document ingestion, preprocessing, model inference, and extraction — improving reliability and reducing manual pipeline monitoring.
- Engineered a production-grade KYC document intelligence pipeline combining document classification with LLM-driven extraction, reducing manual review effort and accelerating verification turnaround.
- Built a data pipeline to ingest and store unstructured KYC documents (PDFs/images) in Azure Data Lake, enabling scalable, secure storage for raw document data.
- Developed RESTful inference services using FastAPI to expose AI/analytics capabilities for enterprise document processing.

- Integrated embedding-based semantic retrieval into LLM workflows to improve contextual understanding and response quality.
- Deployed production-ready services with MLflow-based tracking and version management; containerized components using Docker for consistent deployment.
- Implemented CI/CD pipelines using Azure DevOps to automate application deployment, testing, and release management.
- Collaborated with cross-functional stakeholders in an Agile/Scrum environment to deliver scalable solutions aligned with business requirements.

Evolent, Arlington, VA

Nov 2020 – Jun 2023

Senior Data Scientist / ML Engineer

Project: Enterprise Phishing Detection System (Healthcare) | Tech Stack: Python (Scikit-learn, XGBoost), SQL Server, MLflow, Tableau

- Built and operationalized a phishing detection ML pipeline for a healthcare services organization, using MLflow for full experiment lifecycle management and Tableau to translate model behavior into stakeholder-ready insights.
- Collected, stored, and managed phishing/safe email datasets in SQL Server; wrote SQL queries to extract, clean, and prepare data for analysis.
- Performed data cleaning and feature engineering — URL length, special-character flags, TF-IDF vectors — using Python (Scikit-learn).
- Managed the complete ML lifecycle: data preparation, feature engineering, model training, evaluation, deployment, and monitoring.
- Trained and evaluated classification models (Logistic Regression, XGBoost) using Accuracy, Precision, Recall, and F1-score.
- Used MLflow to track experiments, compare models, and manage versions to select the best-performing model.
- Built interactive Tableau dashboards to visualize model performance, feature importance, and phishing trends for non-technical stakeholders.

Sakav Technology, Pune, MH

Jun 2014 – Feb 2016

Associate Data Scientist

Project: Geospatial Valuation Engine & Real Estate Risk Analytics

Tech Stack: Python, Scikit-learn, XGBoost, CatBoost, SQL, Tableau, SHAP, Statsmodels

Project Overview

Developed a high-fidelity automated valuation model (AVM) to predict residential property values across diverse regional markets. Focused on engineering spatially aware features and ensuring model transparency for stakeholder decision-making, resulting in a reduction in mean absolute percentage error (MAPE) compared to baseline institutional models.

Roles and Responsibilities:

- Engineered interaction terms and geospatial aggregates (e.g., price-per-square-foot trends at the neighborhood level); utilized Recursive Feature Elimination (RFE) and Lasso (L1) regularization to identify the most predictive variables while reducing model complexity.
- Deployed a stacked ensemble of Gradient Boosted Trees (XGBoost/CatBoost) to capture complex non-linear relationships in housing data, significantly outperforming traditional OLS regression in high-volatility markets.
- Implemented a Time-Series Split validation strategy to prevent data leakage, ensuring the model could generalize to future market conditions rather than simply memorizing historical price points.
- Conducted rigorous Multicollinearity diagnostics (VIF) and Hypothesis Testing (p-values) on regression coefficients to ensure the structural integrity of the model and provide reliable feature-importance rankings.
- Integrated SHAP (SHapley Additive exPlanations) to decompose individual house price predictions into contributing factors (e.g., +\$10k for school district, -\$5k for age), providing "Reason Codes" essential for regulatory compliance.
- Designed an automated Anomaly Detection pipeline using Isolation Forests to flag unrealistic property listings or "fat-finger" data entry errors, ensuring data integrity before model training.

Virtuoso Infotech Pvt. Ltd.

Apr 2011 – March 2014

Junior Data Analyst / Database Associate

Project: Enterprise Reporting & Database Support

Tech Stack: SQL Server (T-SQL), SSIS (Integration Services), SSRS (Reporting Services), MS Excel, Python, Jira

Project Overview

Supported the development of a centralized reporting system for vehicle safety data. Focused on automating data entry processes and creating standardized reports to ensure high data quality and accuracy for executive decision-making.

Roles and Responsibilities:

- SSIS Package Development: Created and scheduled automated SSIS packages to move data from text files into SQL Server; moved the team from manual data loading to an automated daily refresh schedule.
- Data Reconciliation & Audit: Conducted thorough Source-to-Target (S2T) audits using SQL comparison scripts; identified and corrected calculation errors that resulted in the recovery of misallocated costs.
- Standardized Reporting (SSRS): Developed a library of automated reports in SSRS, allowing users to filter by date and region, which eliminated the need for manual one-off data requests.

EDUCATION

- Master of Science in Computing and Data Science
- Master of Business Administration

CERTIFICATIONS

- Scrum Master Certified
- CompTIA IT Fundamentals

Authorized to work in the USA (EAD holder). No visa sponsorship required. Open to relocation.